DAY 7 – DATA VISUALIZATION USING GGPLOT2

1. Introduction to Data Visualization

Data Visualization is the process of representing data using charts, graphs and diagrams.

It helps Data Analysts understand data easily and identify patterns, trends and comparisons.

Why do we use Data Visualization?

1. To understand large datasets.
2. To compare sales and performance.
3. To identify trends.
4. To find patterns.
5. To communicate business insights.
6. To prepare reports and dashboards.

Examples:

- Monthly sales comparison – Column Chart
- Sales trends over time – Line Chart
- Relationship between salary and experience – Scatter Plot
- Salary distribution – Histogram
- Identifying outliers – Boxplot

2. Introduction to ggplot2

`ggplot2` is a popular R package used to create professional data visualizations.

It follows a layered approach to create charts.

Step 1: Install package

```
install.packages("ggplot2")
```

Step 2: Load package

```
library(ggplot2)
```

3. Understanding ggplot() Structure

Basic syntax:

```
ggplot(data, aes(x = column1, y = column2)) +
  geom_col()
```

Important Components:

Component	Purpose
`ggplot()`	Starts a chart
`data`	Dataset used
`aes()`	Defines variables and mappings
`geom_*()`	Defines chart type
`labs()`	Adds titles and labels
`theme_*()`	Changes chart appearance

Example:

```
ggplot(sales, aes(x = Month, y = Sales)) +
  geom_col()
```

4. Create Sample Sales Dataset

We will use this dataset for our practical examples.

```
sales <- data.frame(
  Month = c("Jan", "Feb", "Mar", "Apr", "May", "Jun"),
  Sales = c(50000, 65000, 55000, 80000, 75000, 95000)
)

sales
```

5. Column Chart

A Column Chart is used to compare values across different categories.

Example: Month-wise sales comparison.

```
ggplot(sales, aes(x = Month, y = Sales)) +
  geom_col()
```

Add chart title and axis labels:

```
ggplot(sales, aes(x = Month, y = Sales)) +
  geom_col() +
  labs(
    title = "Monthly Sales Analysis",
    x = "Month",
    y = "Total Sales"
  )
```

Business Use: Compare monthly, quarterly or yearly sales.

6. Bar Chart

A Bar Chart represents categories using horizontal bars.

```
ggplot(sales, aes(x = Month, y = Sales)) +
  geom_col() +
  coord_flip()
```

Business Use: Compare product-wise sales or department-wise performance.

7. Line Chart

A Line Chart is used to identify trends over time.

Example: Monthly sales trend.

```
ggplot(sales, aes(x = Month, y = Sales, group = 1)) +
  geom_line() +
```

```
  geom_point()
```

Add title:

```
ggplot(sales, aes(x = Month, y = Sales, group = 1)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Monthly Sales Trend",
    x = "Month",
    y = "Sales"
  )
```

Business Use: Understand whether sales are increasing or decreasing over time.

8. Scatter Plot

A Scatter Plot is used to identify the relationship between two numerical variables.

Example: Experience vs Salary.

Create dataset:

```
employees <- data.frame(
  Experience = c(1, 2, 3, 4, 5, 6, 7, 8),
  Salary = c(25000, 30000, 35000, 45000, 50000, 60000, 70000, 85000)
)
```

Create scatter plot:

```
ggplot(employees, aes(x = Experience, y = Salary)) +
  geom_point()
```

Add title:

```
ggplot(employees, aes(x = Experience, y = Salary)) +
  geom_point() +
  labs(
    title = "Experience vs Salary",
    x = "Experience in Years",
    y = "Salary"
  )
```

Business Use: Understand the relationship between employee experience and salary.

9. Histogram

A Histogram is used to understand the distribution of numerical data.

Example: Salary distribution.

```
ggplot(employees, aes(x = Salary)) +
  geom_histogram()
```

Add title and customize bins:

```
ggplot(employees, aes(x = Salary)) +
  geom_histogram(bins = 5) +
  labs(
    title = "Employee Salary Distribution",
    x = "Salary",
    y = "Frequency"
  )
```

Business Use: Understand how salary values are distributed among employees.

10. Boxplot

A Boxplot is used to understand the distribution of data and identify potential outliers.

Example:

```
ggplot(employees, aes(y = Salary)) +
  geom_boxplot()
```

Add title:

```
ggplot(employees, aes(y = Salary)) +
  geom_boxplot() +
  labs(
    title = "Salary Distribution",
    y = "Salary"
  )
```

Business Use: Identify unusually high or low salary values.

11. Adding Colours to Charts

We can customize charts using colour.

Example:

```
ggplot(sales, aes(x = Month, y = Sales)) +
  geom_col(fill = "steelblue")
```

Different colour by category:

```
ggplot(sales, aes(x = Month, y = Sales, fill = Month)) +
  geom_col()
```

12. Adding Data Labels

We can display values directly on a chart using `geom_text()`.

Example:

```
ggplot(sales, aes(x = Month, y = Sales)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = Sales), vjust = -0.5) +
  labs(
    title = "Monthly Sales with Data Labels",
    x = "Month",
    y = "Sales"
  )
```

13. Changing Chart Theme

Themes are used to change the overall appearance of charts.

Example 1:

```
ggplot(sales, aes(x = Month, y = Sales)) +
  geom_col() +
  theme_minimal()
```

Example 2:

```
ggplot(sales, aes(x = Month, y = Sales)) +
  geom_col() +
  theme_classic()
```

Example 3:

```
ggplot(sales, aes(x = Month, y = Sales)) +
  geom_col() +
  theme_bw()
```

14. Saving Charts

We can save charts as image files using `ggsave()`.

Syntax:

```
ggsave("filename.png", width = 8, height = 5)
```

Example:

```
ggsave(
  "Monthly_Sales_Chart.png",
  width = 8,
  height = 5
)
```

The chart is saved in the current working directory.

Check the folder:

```
getwd()
```

15. Complete Practical – Employee Data Visualization

Create the following dataset:

```
employees <- data.frame(
  Name = c("Amit", "Rahul", "Priya", "Neha", "Vikas", "Sneha", "Karan",
"Pooja"),
  Department = c("Sales", "IT", "Sales", "HR", "IT", "Sales", "HR",
"IT"),
  Experience = c(1, 2, 3, 4, 5, 6, 7, 8),
  Salary = c(25000, 30000, 35000, 45000, 50000, 60000, 70000, 85000)
)
```

Task 1: Salary Column Chart

```
ggplot(employees, aes(x = Name, y = Salary)) +
  geom_col() +
  labs(title = "Employee-wise Salary")
```

Task 2: Department-wise Employee Count

```
ggplot(employees, aes(x = Department)) +
  geom_bar() +
```

```
  labs(title = "Department-wise Employee Count")
```

Task 3: Experience vs Salary

```
ggplot(employees, aes(x = Experience, y = Salary)) +
  geom_point() +
  labs(title = "Experience vs Salary")
```

Task 4: Salary Distribution

```
ggplot(employees, aes(x = Salary)) +
  geom_histogram(bins = 5) +
  labs(title = "Salary Distribution")
```

Task 5: Salary Boxplot

```
ggplot(employees, aes(y = Salary)) +
  geom_boxplot() +
  labs(title = "Employee Salary Boxplot")
```

16. Student Practice Questions

Use the sales dataset and complete the following:

1. Create a monthly sales Column Chart.
2. Create a monthly sales Line Chart.
3. Add chart title and axis labels.
4. Change the colour of the Column Chart.
5. Display sales values as data labels.
6. Create a horizontal Bar Chart.
7. Create an employee salary Histogram.
8. Create an Experience vs Salary Scatter Plot.
9. Create a Salary Boxplot.
10. Save any two charts as PNG files.

17. Homework Assignment – Sales Visualization

Create the following dataset:

```
sales <- data.frame(
  Product = c("Laptop", "Mouse", "Keyboard", "Monitor", "Laptop",
"Mouse"),
  Region = c("West", "North", "South", "West", "East", "North"),
```

```
  Quantity = c(2, 10, 5, 3, 1, 8),
  Sales = c(120000, 10000, 15000, 60000, 65000, 8000)
)
```

Perform the following:

1. Create a Product-wise Sales Column Chart.
2. Create a Region-wise Sales Bar Chart.
3. Create a Quantity vs Sales Scatter Plot.
4. Create a Sales Histogram.
5. Add chart titles and axis labels.
6. Customize the chart theme.
7. Display data labels on the sales chart.
8. Export charts as PNG images.

Expected Output: A collection of professional sales charts created using ggplot2.

18. Important Functions – Day 7

Function	Purpose
`ggplot()`	Create chart
`aes()`	Map variables
`geom_col()`	Column chart
`geom_bar()`	Bar chart using counts
`coord_flip()`	Horizontal chart
`geom_line()`	Line chart

`geom_point()`	Scatter plot
`geom_histogram()`	Histogram
`geom_boxplot()`	Boxplot
`geom_text()`	Add data labels
`labs()`	Add titles and labels
`theme_minimal()`	Minimal theme
`theme_classic()`	Classic theme
`theme_bw()`	Black and white theme
`ggsave()`	Save chart

19. Data Visualization Workflow

IMPORT DATA

↓

SELECT VARIABLES

↓

CHOOSE CHART TYPE

↓

CREATE VISUALIZATION

↓

ADD TITLE AND LABELS

↓

CUSTOMIZE CHART

↓

SAVE CHART

↓

INTERPRET INSIGHTS

Remember: A good Data Analyst does not just create charts. A good Data Analyst explains what the chart tells us about the business.